

Tianyou Lai

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Machine Learning Engineering

Carnegie Mellon master's student with experience in model development, dataset preparation, and experimental evaluation across computer vision, LLMs, and time-series forecasting. Combines published ML research with Python/SQL data engineering and distributed-data infrastructure experience.

Education

Carnegie Mellon University M.S. in Artificial Intelligence Systems Management	Aug. 2026 – Dec. 2027 (expected)
Lanzhou University B.S. in Data Science & Big Data Technology GPA: 87.34/100	Sep. 2021 – Jul. 2025
The University of Texas at Austin Software Engineering Exchange Program GPA: 4.0/4.0	Jan. 2024 – Feb. 2024

Technical Skills

Programming and AI: Python, SQL, C/C++; PyTorch, TensorFlow; object detection, LLM fine-tuning and evaluation, federated learning, time-series modeling.

Data and infrastructure: MySQL, Hadoop, Spark, OpenStack, Docker, Kubernetes, Linux.

Evaluation and communication: Benchmarking, ablation studies, cross-domain evaluation; Matplotlib, Seaborn, Tableau, PowerPoint.

Selected Machine Learning Projects

Reasoning after Unlearning | LLM Evaluation 2026
First author | Submitted to ICLR 2027

- Designed controlled TOFU evaluations across original, unlearned, and retrained Llama checkpoints to distinguish context effects, answer exposure, and answer-information use.
- Compared direct answering and matched-length reasoning controls; used answer masking and fixed-input identity substitution to probe evaluation failure modes.

TRACE | Interpretable Breast Ultrasound Classification Feb. 2026 – Present
Co-first author | Southern Medical University collaboration | ACM MM 2026, accepted

- Contributed to a concept-based diagnostic framework that uses radiology reports for training supervision and requires only images at inference.
- Worked on BI-RADS-aligned image recognition and structured clinical descriptions using private and public-derived datasets; the framework distills concept edits into an image-only self-editor.

FREDNet | Industrial Defect Detection Sep. 2024 – Jul. 2025
Co-first author | ICASSP 2026

- Designed and implemented frequency and decomposed-spatial modules with hierarchical attention fusion for surface-defect detection.
- Led NEU-DET and GC-10 data curation, training, benchmarking, ablations, and cross-domain evaluation; reported a 4.8% improvement in mAP@0.5.

ReLaMix | Forecasting with Delayed Observations Jan. 2026
Co-first author | arXiv preprint, 2026

- Developed residual bottleneck mixing for stale financial observations; evaluated second-resolution PAXGUSDT forecasts and cross-asset transfer to BTCUSDT.
- Used non-overlapping temporal segments to prevent evaluation leakage; benchmarked delay robustness and model size against mixer and Transformer baselines.

Engineering and Industry Experience

Sichuan DeepGraph Intelligent Technology

Sep. 2024 – Feb. 2025

Data Science Intern | Chengdu, China

- Prepared datasets in Python and validated integrity and consistency for downstream analysis and modeling.
- Queried graph relationships with Ultpa; supported analysis-algorithm optimization and OpenStack deployment.

Lanzhou Bronze Ding Intelligent Technology

Mar. 2024 – Sep. 2024

Data Engineer Intern | Lanzhou, China

- Processed 20+ datasets and labeled 10,000+ records with Python and SQL; produced seven analytical reports using Matplotlib and Seaborn.
- Deployed MySQL and Hadoop/Spark on a private cloud to support distributed data storage and processing.

Guotai Junan Futures

Mar. 2024 – Apr. 2024

Quantitative Research Assistant | Remote

- Extended a Python futures trading system with position locking, closing, and batch-order functions across products and exchanges.
- Debugged C++/Python CTP-API integrations in Simnow; implemented TWAP/VWAP execution classes and TA-Lib indicator generation and filtering.

Zhongzheng Tiantong CPA

Jul. 2023 – Sep. 2023

Tax and Business Consulting Intern | Beijing, China

- Cleaned financial data with Python/SQL and created Excel/Tableau analyses for client financial, risk, and market-trend reports.

Feitian Technologies

Jul. 2022 – Jan. 2023

Product Data Engineer Intern | Beijing, China

- Maintained hardware-product data with Python/SQL; implemented and tested password-system features and reviewed C/C++ code.
- Collaborated on 30+ performance tests and improvement plans; delivered 10+ technical and industry research reports.

Additional Research

- **LLM fine-tuning:** Independent TCM honors thesis; curated data and developed prompting/fine-tuning workflows, reducing text length up to 44.1%, increasing terminology tokens 4.4-fold, and reaching diagnostic accuracy up to 81.3%.
- **Federated ML and model behavior:** Led a published heterogeneity survey and developed frequency-adaptive semantic distillation. Co-authored AAAI/ICLR 2027 submissions on heterogeneous edge learning, federated language/vision-language models, multimodal unlearning, model selection, and multi-agent cooperation.
- **Human-AI interaction:** First author of a paper submitted to CHI 2027. Distinguished agreement with an LLM judgment, acceptance of its authority, and subsequent action in livestream games and academic reviewing; used observational, interview, and presentation-comparison methods to study source attribution and human mediation.
- **Scientific ML:** Built metasurface regression (mean MSE 0.00331), wind forecasting (best R^2 0.9905), and Spike/EEG detection studies. Preparing a CHF-AI review for *Energy*; auditing a public benchmark and connecting validation under distribution shift, uncertainty, and deployment constraints to engineering-use boundaries.

Selected Publications and Honors

ACM MM 2026 (TRACE*, accepted); ICASSP 2026 (FREDNet*); arXiv 2026 (ReLaMix*); Expert Systems 2026 (federated learning survey). Full list: laityy.github.io/publications/.

* Tianyou Lai is a co-first author.

Lanzhou University academic, internship, and international exchange scholarships; undergraduate honors thesis (2025). Mandarin (native); English (TOEFL 103).